

GW2025 - HW5

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Question 1

- Event name: GRB 200826A
- Location: 4.67(RA), 35.29(DEC), $z = 0.7481$
- Trigger time (UTC): 2020-08-26 04:29:52.571
- T_{90} : 1.136

This burst is unique in that it might originate from a massive core collapse event though its duration (T_{90}) is quite short (< 2 s), which poses a challenge to the dominant classification paradigms for GRBs. The burst was detected by Fermi-GBM, and the data from three detectors (nb, n7 and b1) below are collected from [NJU Fermi pipeline](#). We could find that according to the values for T_{90} , it belongs to a typical short burst. While its position on the $E_p - E_{\text{iso}}$ map is more closed to the amati relation for long bursts, indicating that they might share a common physical origin.

Question 2

$$\Delta t_{\text{obs}} = \Delta t_e (1 - \beta \cos \theta), \quad (1)$$

$$t_{\text{obs}} = t_e (1 - \beta \cos \theta), \quad (2)$$

$$r(\theta, t_{\text{obs}}) = \beta c t_e = \frac{\beta c t_{\text{obs}}}{1 - \beta \cos \theta}. \quad (3)$$

Thus the equation for Equal Arrival Time Surface (EATS) is obtained, and the surface in x-y plane is shown in Figure 4.

References

Zhang, B.-B., Liu, Z.-K., Peng, Z.-K., Li, Y., Lü, H.-J., Yang, J., Yang, Y.-S., Yang, Y.-H., Meng, Y.-Z., Zou, J.-H., Ye, H.-Y., Wang, X.-G., Mao, J.-R., Zhao, X.-H., Bai, J.-M., Castro-Tirado,

A. J., Hu, Y.-D., Dai, Z.-G., Liang, E.-W., and Zhang, B. (2021). A peculiarly short-duration gamma-ray burst from massive star core collapse. *Nature Astronomy*, 5:911–916.

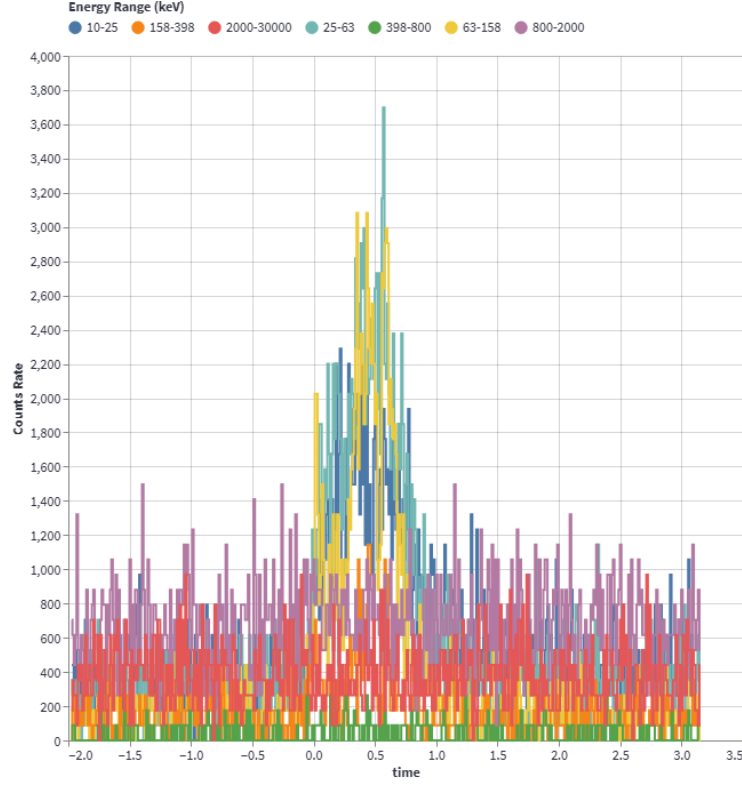


Figure 1: Light curves for different energy bands.

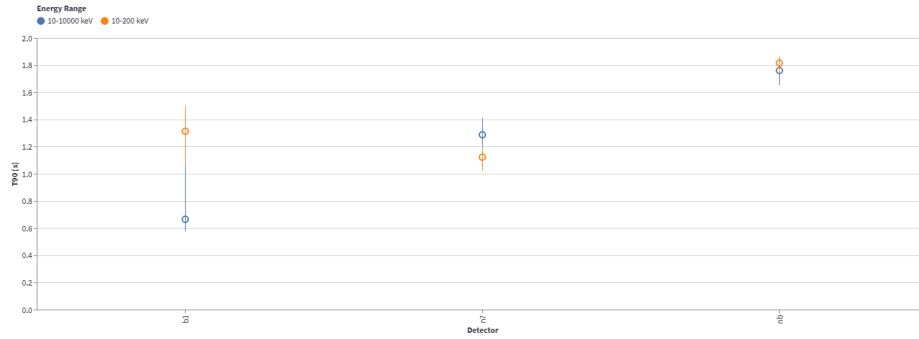


Figure 2: T_{90} of the burst for two distinct energy ranges.

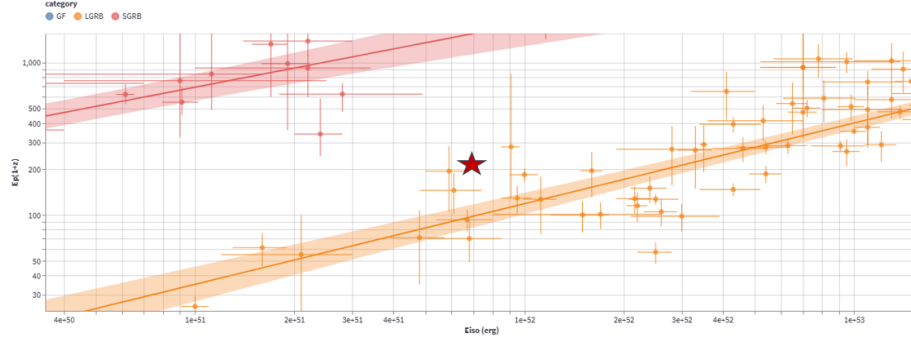


Figure 3: Position of GRB 200826A on the $E_p - E_{\text{iso}}$ map, which is denoted by the red star (from [Zhang et al. \(2021\)](#)).

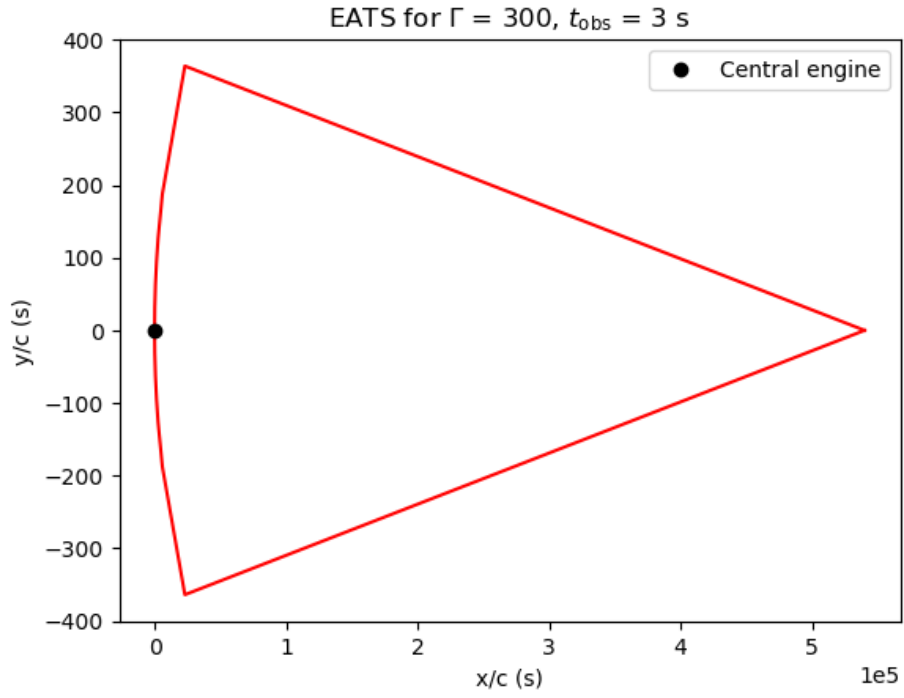


Figure 4: Equal Arrival Time Surface for $\Gamma = 300$, $t_{\text{obs}} = 3$ s.